

# REQUIREMENT ANALYSIS

POCKETSMART AI — Smart Budget & Recommendation Assistant

Expanded phase evidence with detailed wording, diagrams, workflow, and project-focused explanation.

## Understanding the user requirements

Requirement analysis translates the PocketSmart AI idea into concrete application behaviour. The system needs to collect enough information for each planning category without making the user process unnecessary complexity. A common requirement is budget input; domain-specific requirements then refine the recommendation context. Authentication and History are cross-cutting requirements because the application is designed for authenticated users.

## Functional requirements

The documented workflow requires registration and login, a dashboard, dedicated Home/Party/Jewelry planners, recommendation generation, structured result pages, and a History module. The Home planner captures interior-related quantities and room information. The Party planner captures event and guest-related information. The Jewelry planner captures occasion and style preferences and can accept an optional outfit image. These requirements define what the application must allow a user to do.

## Quality and security requirements

The project also has non-functional expectations: protected routes, secure password handling, HTTP-only cookie storage for access tokens, environment-based protection for the Gemini API key, input validation, and user-specific History association. The analysis therefore covers not only visible screens but also the reliability and protection needed behind those screens.

## Visual Model



Figure — Phase visual model

## Process Flow

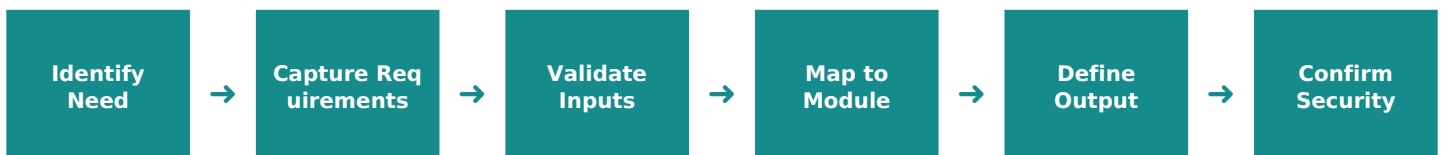


Figure — Detailed workflow / roadmap

# PHASE 2 — DETAILED CONTENT

PocketSmart AI connects practical planning with a structured recommendation workflow. The following section expands the phase in clear project language so the document has sufficient explanatory depth for academic/project review.

## Project Explanation

A strong budget-planning assistant should reduce uncertainty without hiding the reasoning context behind the result. In PocketSmart AI, the user supplies the information that matters for the selected category, including budget and relevant preferences. The application then processes that context and presents a structured recommendation. This structure makes the output easier to read, compare, and explain during a project demonstration. The project also demonstrates how a common architecture can support different use cases. Home Interior planning focuses on rooms, furniture, lighting, fans, and dining requirements. Party planning focuses on guests, event type, venue type, catering, decoration, and entertainment. Jewelry planning focuses on occasion, style preferences, budget, and optional outfit-image context. Although the inputs differ, the overall application pattern remains consistent.

## Key Concepts

| Requirement    | PocketSmart AI interpretation                                       |
|----------------|---------------------------------------------------------------------|
| Authentication | Registration, login, protected access, logout.                      |
| Planning       | Home Interior, Party, and Jewelry planner inputs.                   |
| Recommendation | Gemini-based personalized, budget-aware output.                     |
| History        | Saved recommendation records for authenticated users.               |
| Security       | Password hashing, protected tokens, API-key protection, validation. |

## Roadmap / Mind-Map View

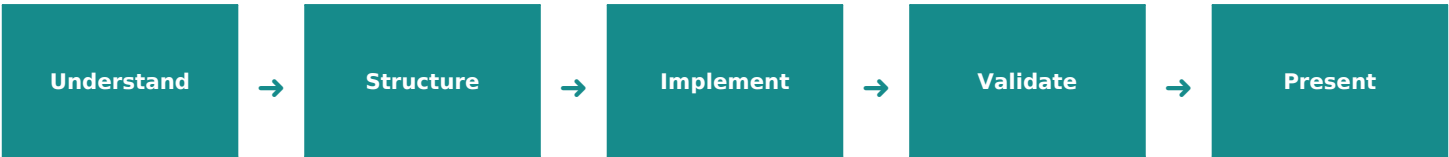


Figure — Additional project roadmap visual

## Project Evidence

The supplied project documentation includes interface and workflow evidence. A representative screenshot is included here to connect the written explanation with the implemented application.

Home Interior Budget Planner

Create a customized budget plan for your dream home interior

1 Budget Details

Total Budget (\$)

25000

2 Fixtures & Furniture

Number of Lights/Fixtures

8

Number of Ceiling Fans

6

Number of Furniture Pieces

4

Number of Dining Tables

2

3 Rooms to Include

☒ Living Room

☒ Kitchen

☒ Bedroom

4 Additional Information

Special Requirements or Preferences

Any specific requirements or preferences...

Generate Recommendations

Figure — Representative PocketSmart AI project evidence

# PHASE SUMMARY

Phase 2 contributes to the overall PocketSmart AI lifecycle by making the project easier to understand, build, validate, or present. The emphasis in this phase is on clear communication: the reader should be able to see what the system is doing, why the step matters, and how it connects to the final budget-aware recommendation workflow.

## End-to-End Connection



## Project Reflection

PocketSmart AI is designed around a practical user problem: making planning decisions while respecting a defined budget. The project brings together web development, authentication, structured data, Generative AI, visual presentation, and History. This combination makes the application a useful example of how a software project can move from a simple idea to a demonstrable end-to-end system. The documentation should therefore communicate not only the features, but also the reasoning, workflow, and evidence behind them.

## Closing Note

The phase is complete when its purpose, workflow, supporting evidence, and connection to PocketSmart AI are understandable to a reviewer without requiring them to infer missing steps. The expanded paragraphs and diagrams in this version are intended to make that explanation clearer and more professional.